24CS411 DATABASE MANAGEMENT SYSTEMS LABORATORY

1. Database Design using ER modeling, normalization and Implementation for any application

2. Data Definition Commands, Data Manipulation Commands for inserting, deleting, updating and retrieving Tables

3. Integrity Constraints

4. SQL Functions – Single Row Functions (Numeric, String, Date) and Aggregate Functions Group by clause, Having Clause.

5. Database Querying – Simple queries, Nested queries, Sub queries and Joins

6. Views, Sequences, Synonyms

7. Data Control and Transaction Control statements

8. Database Programming: Implicit and Explicit Cursors

9. Procedures and Functions

10. Triggers

11. Database Connectivity with Front End Tools

12. Simple Queries in MongoDB

Ex 02: **Data Definition Commands and Data Manipulation Commands**

AIM: To practice basic SQL DDL Commands like Create, Alter, Drop and Truncate and DML Commands like Insert, update, delete and select.

Introduction:

SQL (pronounced "ess-que-el") stands for Structured Query Language. SQL is used to communicate with a database. SQL statements are used to perform tasks such as update or retrieve data from a database. Some common relational database management systems are: Oracle, Sybase, Microsoft SQL Server, Access, Ingres, etc. Although most database systems use SQL.

Data Types.

DATA TYPE	SPECIFICATION	DESCRIPTION
Char	CHAR(size)	Character Data. Size is specified in number.
Varchar2	Varchar2(size)	Same as above.(Recommended for less storage)
Date	Date	Used for specifying dates
Number	Number	Used to specify numeric data. Number (size, dec) with digits equal to specified size. Number (5, 2) can store up to 999.99.
BLOB	For storing picture information	Binary Large Object
Integer	Number	Same as number
Float	Number	Same as number

DDL[Data Definition Language]

DDL is concerned with the creation, deletion and modification of database objects like tables, indexes and views.

CREATE - to create objects in the database

ALTER - alters the structure of the database

DROP - delete objects from the database

TRUNCATE - remove all records from a table, including all spaces allocated for the records are removed

RENAME - rename an object

1. The Create Table Command: - it defines each column of the table uniquely. Each column has minimum of three attributes, a name , data type and size.

Syntax:

Create table <table name> (<col1> <datatype>(<size>),<col2> <datatype><size>));

2. Modifying the structure of tables.

a)add new columns

Syntax:

Alter table <tablename> add(<new col><datatype(size),<new col>datatype(size));

3. Dropping a column from a table.

Syntax:

Alter table <tablename> drop column <col>;

4. Modifying existing columns.

Syntax:

Alter table <tablename> modify(<col><newdatatype>(<newsize>));

5. Renaming the tables

Syntax:

Rename <oldtable> to <new table>;

6. truncating the tables.

Syntax:

Truncate table <tablename>;

7. Destroying tables.

Syntax:

Drop table <tablename>;

DATA MANIPULATION LANGUAGE

The Data Manipulation Language (DML) is used to retrieve, insert and modify database information. The basic DML commands include

INSERT

SELECT

UPDATE

DELETE

1. Inserting Data into Tables:

Syntax:

insert into <tablename> (<col1>,<col2>) values(<exp>,<exp>);

2. Delete operations

a) remove all rows

Syntax:

delete from <tablename>;

b) removal of a specified row/s

Syntax:

delete from <tablename> where <condition>;

3. Updating the contents of a table.

a) updating all rows

Syntax:

update <tablename> set <col>=<exp>,<col>=<exp>;

b) updating seleted records.

Syntax:

update <tablename> set <col>=<exp>,<col>=<exp> where <condition>;

4. Viewing data in the tables:

a) all rows and all columns

Syntax:

Select <col> to <col n> from tablename;

5. Filtering table data:

a) Selected columns and all rows:

Syntax:

select <col1>,<col2> from <tablename>;

b) selected rows and all columns:

Syntax:

select * from <tablename> where <condition>;

c) selected columns and selected rows

Syntax:

select <col1>,<col2> from <tablename> where<condition>;

6. Sorting data in a table.

Syntax:

select * from <tablename> order by <col1>,<col2> <[sortorder]>;

Result:

Thus the various DDL commands and DML commands are studied and the results are verified.

Ex No. 04 **SQL Functions – Single Row Functions (Numeric, String, Date) and Aggregate Functions Group by clause, Having Clause**

Aim:

To study the various operators and built-in functions in SQL for Oracle.

SQL OPERATORS

Arithmetic Operators	Description
+,-,*,/	To perform basic arithmetic operations.

Comparison Operators	**Description**
=	Equal to
<>, !=	is not equal to
<	less than
>	greater than
>=	greater than or equal to
<=	less than or equal to
LIKE	column value is similar to specified character(s).
IN	column value is equal to any one of a specified set of values.
BETWEEN...AND	column value is between two values, including the end values specified in the range.
IS NULL	column value does not exist.
Logical Operators	**Description**
OR	For the row to be selected at least one of the conditions must be true.
AND	For a row to be selected all the specified conditions must be true.

NOT	For a row to be selected the specified condition must be false.

SQL FUNCTIONS

There are two types of built in functions in Oracle.

1) **Single Row Functions**: Single row or Scalar functions return a value for every row that is processed in a query.

2) **Group Functions:** These functions group the rows of data based on the values returned by the query. This is discussed in SQL GROUP Functions. The group functions are used to calculate aggregate values like total or average, which return just one total or one average value after processing a group of rows

There are four types of single row functions.
They are:

1) **Numeric Functions:** These are functions that accept numeric input and return numeric values.

2) **Character or Text Functions:** These are functions that accept character input and can return both character and number values.

3**) Date Functions**: These are functions that take values that are of datatype DATE as input and return values of datatype DATE, except for the MONTHS_BETWEEN function, which returns a number.

4) **Conversion Functions:** These are functions that help us to convert a value in one form to another form. For Example: a null value into an actual value, or a value from one datatype to another datatype like NVL, TO_CHAR, TO_NUMBER, TO_DATE etc..,

Use of DUAL Table in Oracle

This is a single row and single column dummy table provided by oracle. This is used to perform mathematical calculations without using a table.

Numeric Functions:

Numeric functions are used to perform operations on numbers. They accept numeric values as input and return numeric values as output. Few of the Numeric functions are:

Function Name	**Return Value**
ABS (x)	Absolute value of the number '*x*'
CEIL (x)	Integer value that is Greater than or equal to the number '*x*'
FLOOR (x)	Integer value that is Less than or equal to the number '*x*'
TRUNC (x, y)	Truncates value of number '*x*' up to '*y*' decimal places
ROUND (x, y)	Rounded off value of the number '*x*' up to the number '*y*' decimal places

Character or Text Functions: Character or text functions are used to manipulate text strings. They accept strings or characters as input and can return both character and number values as output.

Few of the character or text functions are as given below:

Function Name	**Return Value**
LOWER (string_value)	All the letters in *'string_value'* is converted to lowercase.
UPPER (string_value)	All the letters in *'string_value'* is converted to uppercase.
INITCAP (string_value)	All the letters in *'string_value'* is converted to mixed case.
LTRIM (string_value, trim_text)	All occurrences of *'trim_text'* is removed from the left of *'string_value'*.
RTRIM (string_value, trim_text)	All occurrences of *'trim_text'* is removed from the right of *'string_value'* .
TRIM (trim_text FROM string_value)	All occurrences of *'trim_text'* from the left and right of *'string_value'* , *'trim_text'* can also be only one character long .
SUBSTR (string_value, m, n)	Returns *'n'* number of characters from *'string_value'* starting from the '*m'* position.
LENGTH (string_value)	Number of characters in *'string_value'* in returned.

Date Functions:

These are functions that take values that are of datatype DATE as input and return values of datatypes DATE, except for the MONTHS_BETWEEN function, which returns a number as output.

Function Name	**Return Value**
ADD_MONTHS (date, n)	Returns a date value after adding *'n'* months to the date *'x'*.
MONTHS_BETWEEN (x1, x2)	Returns the number of months between dates x1 and x2.
NEXT_DAY (x, week_day)	Returns the next date of the *'week_day'* on or after the date *'x'* occurs.
LAST_DAY (x)	It is used to determine the number of days remaining in a month from the date *'x'* specified.
SYSDATE	Returns the systems current date and time.

Conversion Functions:

These are functions that help us to convert a value in one form to another form. For Ex: a null value into an actual value, or a value from one datatype to another datatype like NVL, TO_CHAR, TO_NUMBER, TO_DATE.

Function Name	Return Value
TO_CHAR (x [,y])	Converts Numeric and Date values to a character string value. It cannot be used for calculations since it is a string value.
TO_DATE (x [, date_format])	Converts a valid Numeric and Character values to a Date value. Date is formatted to the format specified by 'date_format'.
DECODE (a, b, c, d, e, default_value)	Checks the value of 'a', if a = b, then returns 'c'. If a = d, then returns 'e'. Else, returns default_value.

<u>Aggregate Functions</u>

An aggregate function is a function that performs a calculation on a set of values, and returns a single value.

Aggregate functions are often used with the GROUP BY clause of the SELECT statement. The GROUP BY clause splits the result-set into groups of values and the aggregate function can be used to return a single value for each group.

The most commonly used SQL aggregate functions are:

- MIN() - returns the smallest value within the selected column
- MAX() - returns the largest value within the selected column
- COUNT() - returns the number of rows in a set
- SUM() - returns the total sum of a numerical column
- AVG() - returns the average value of a numerical column

SQL GROUP BY Clause

The SQL GROUP BY Clause is used along with the group functions to retrieve data grouped according to one or more columns.

Group functions are built-in SQL functions that operate on groups of rows and return one value for the entire group. These functions are: COUNT, MAX, MIN, AVG, SUM, DISTINCT

COUNT (): This function returns the number of rows in the table that satisfies the condition specified in the WHERE condition. If the WHERE condition is not specified, then the query returns the total number of rows in the table.

DISTINCT(): This function is used to select the distinct rows.

MAX(): This function is used to get the maximum value from a column.

MIN(): This function is used to get the minimum value from a column.

AVG(): This function is used to get the average value of a numeric column.

SUM(): This function is used to get the sum of a numeric column.

HAVING Clause

The SQL HAVING clause filters the results of grouped data after using the GROUP BY clause. It is used with aggregate functions such as SUM(), COUNT(), or AVG() to display only those groups that meet specific conditions.

Result: Thus the various SQL functions are studied and the results are verified.

Ex. No: 7 **Data Control and Transaction Control statements**

Aim:

To execute DCL and TCL commands.

INTRODUCTION

DCL COMMANDS

DCL stands for **Data Control Language** in Structured Query Language (SQL). These commands are used to control privileges in the database. The privileges (right to access the data) are required for performing all database operations like creating tables, views, or sequences.

Need of DCL Commands

- Unauthorized access to the data is prevented to achieve security in database.
- DCL commands maintain the database effectively; no one other than the database administrator is allowed to access the data without permission.
- These commands provide flexibility to the data administrator to set and remove database permissions in a granular fashion.

Commands in DCL

The two most important DCL commands are:

1. GRANT

This command is used to grant permission to the user to perform a particular operation on a particular object. Database Administrator can grant privileges according to the requirement.

Syntax:

GRANT privilege_list ON object_name TO user_name;

2. REVOKE

This command is used to take permission or access back from the user. If the administrator wants to remove the permissions that were previously granted, the REVOKE command is used.

Syntax:

REVOKE privilege_list ON object_name FROM user_name;

TCL COMMANDS

COMMIT Command

The COMMIT command is used to permanently save any transaction into the database. It marks the changes as permanent.

Syntax:

COMMIT;

ROLLBACK Command

This command restores the database to the last committed state. It is also used with the SAVEPOINT command to jump to a save point in an ongoing transaction.

Syntax:

ROLLBACK TO savepoint_name;

SAVEPOINT Command

The SAVEPOINT command is used to temporarily save a transaction so that roll back to that point can be done whenever required.

Syntax:

SAVEPOINT savepoint_name;

EXAMPLES

GRANT Command
GRANT SELECT, UPDATE ON MY_TABLE TO SOME_USER, ANOTHER_USER;

REVOKE Command
REVOKE SELECT, UPDATE ON MY_TABLE FROM USER1, USER2;

RESULT

Thus, the DCL and TCL commands were executed successfully.